

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.

```
from google.colab import files

uploaded = files.upload()
```

Choose Files c107fa55-4...8d1fb0c.csv
c107fa55-45be-4f9c-8f61-088e88d1fb0c.csv(text/csv) - 26651 bytes, last modified: 6/26/2026 - 100% done
Saving c107fa55-45be-4f9c-8f61-088e88d1fb0c.csv to c107fa55-45be-4f9c-8f61-088e88d1fb0c (1).csv

MULTIPLE LINEAR REGRESSION - MULTI-CHANNEL MARKETING ANALYSIS.

PROJECT OVERVIEW

This project analyzes how different marketing channels influence sales using Multiple Linear Regression. The objective is to identify the contribution of TV advertising, Radio advertising, Social Media advertising, and Influencer marketing to Sales and provide data-driven recommendations for marketing budget allocation.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm

from statsmodels.stats.outliers_influence import variance_inflation_factor
```

```
df=pd.read_csv("c107fa55-45be-4f9c-8f61-088e88d1fb0c.csv")

df.head()
```

	TV	Radio	Social Media	Influencer	Sales
0	Low	3.518070	2.293790	Micro	55.261284
1	Low	7.756876	2.572287	Mega	67.574904
2	High	20.348988	1.227180	Micro	272.250108
3	Medium	20.108487	2.728374	Mega	195.102176
4	High	31.653200	7.776978	Nano	273.960377

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    TV          572 non-null    object
1    Radio       572 non-null    float64
2    Social Media 572 non-null    float64
3    Influencer  572 non-null    object
4    Sales       572 non-null    float64
dtypes: float64(3), object(2)
memory usage: 22.5+ KB
```

df.describe()

	Radio	Social Media	Sales
count	572.000000	572.000000	572.000000
mean	17.520616	3.333803	189.296908
std	9.290933	2.238378	89.871581
min	0.109106	0.000031	33.509810
25%	10.699556	1.585549	118.718722
50%	17.149517	3.150111	184.005362
75%	24.606396	4.730408	264.500118
max	42.271579	11.403625	357.788195

```
df.isnull().sum()
```

```

0
TV      0
Radio   0
Social Media  0
Influencer  0
Sales   0

```

dtype: int64

```
df = df.dropna()
```

```
df = pd.get_dummies(df, columns=['TV', 'Influencer'], drop_first=True)
```

```
df.head()
```

	Radio	Social Media	Sales	TV_Low	TV_Medium	Influencer_Mega	Influencer_Micro	Influencer_Nano
0	3.518070	2.293790	55.261284	True	False	False	True	False
1	7.756876	2.572287	67.574904	True	False	True	False	False
2	20.348988	1.227180	272.250108	False	False	False	True	False
3	20.108487	2.728374	195.102176	False	True	True	False	False
4	31.653200	7.776978	273.960377	False	False	False	False	True

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
corr = df.corr(numeric_only=True)
corr
```

	Radio	Social Media	Sales	TV_Low	TV_Medium	Influencer_Mega	Influencer_Micro	Influencer_Nano
Radio	1.000000	0.629941	0.858036	-0.674203	0.004400	0.035704	0.001712	0.015640
Social Media	0.629941	1.000000	0.542048	-0.423910	-0.008272	0.066023	-0.052077	0.028030
Sales	0.858036	0.542048	1.000000	-0.805896	0.050449	0.032443	-0.006503	0.017656
TV_Low	-0.674203	-0.423910	-0.805896	1.000000	-0.550117	-0.052697	0.016107	0.006815
TV_Medium	0.004400	-0.008272	0.050449	-0.550117	1.000000	0.067488	0.007302	-0.059374
Influencer_Mega	0.035704	0.066023	0.032443	-0.052697	0.067488	1.000000	-0.336096	-0.345177
Influencer_Micro	0.001712	-0.052077	-0.006503	0.016107	0.007302	-0.336096	1.000000	-0.368361
Influencer_Nano	0.015640	0.028030	0.017656	0.006815	-0.059374	-0.345177	-0.368361	1.000000

Next steps: [Generate code with corr](#) [New interactive sheet](#)

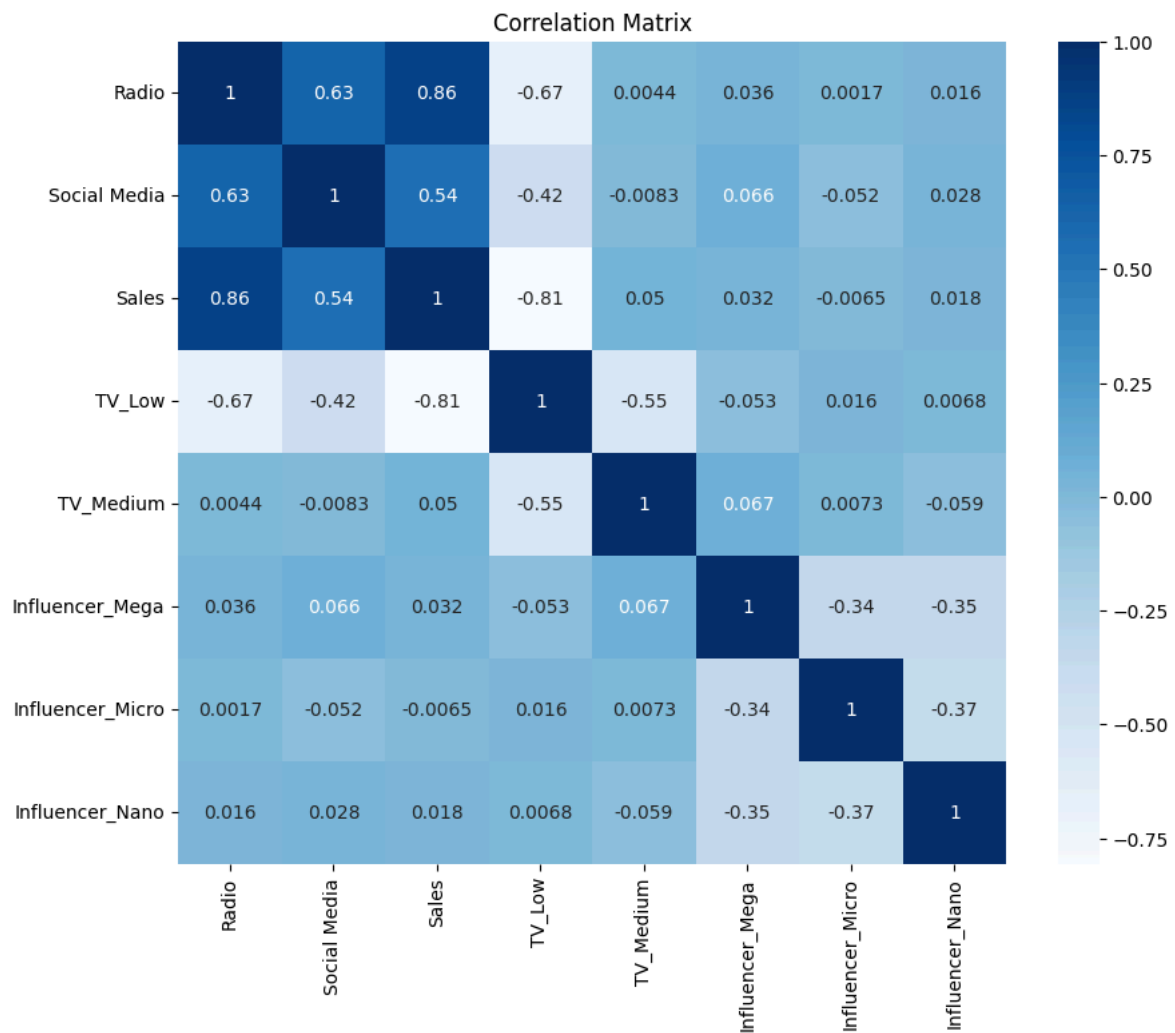
```

plt.figure(figsize=(10,8))

sns.heatmap(corr,
            annot=True,
            cmap="Blues")

plt.title("Correlation Matrix")
plt.show()

```



```
print(df.dtypes)
```

```
Radio          float64
Social Media   float64
Sales          float64
TV_Low         bool
TV_Medium      bool
Influencer_Mega bool
Influencer_Micro bool
Influencer_Nano bool
dtype: object
```

```
print(df.columns)
```

```
Index(['Radio', 'Social Media', 'Sales', 'TV_Low', 'TV_Medium',
      'Influencer_Mega', 'Influencer_Micro', 'Influencer_Nano'],
      dtype='object')
```

```
bool_cols = df.select_dtypes(include='bool').columns
df[bool_cols] = df[bool_cols].astype(int)
```

```
x = df.drop("Sales", axis=1)

x = sm.add_constant(x)

vif = pd.DataFrame()

vif["Variable"] = x.columns

vif["VIF"] = [variance_inflation_factor(x.values, i)
              for i in range(x.shape[1])]

print(vif)
```

```
Variable      VIF
0      const  31.649565
1      Radio   3.479641
2  Social Media  1.669098
3    TV_Low    4.076384
```

```

4     TV_Medium    2.233350
5  Influencer_Mega 1.593889
6  Influencer_Micro 1.618434
7  Influencer_Nano 1.627430

```

```

x = df.drop("Sales", axis=1)

y = df["Sales"]

x = sm.add_constant(x)

model = sm.OLS(y,x).fit()

print(model.summary())

```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          Sales    R-squared:                0.904
Model:                  OLS      Adj. R-squared:            0.903
Method:                 Least Squares    F-statistic:           760.4
Date:                   Sat, 27 Jun 2026    Prob (F-statistic):     1.82e-282
Time:                   15:46:19    Log-Likelihood:        -2713.4
No. Observations:       572        AIC:                   5443.
Df Residuals:           564        BIC:                   5478.
Df Model:                7
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	217.4784	6.584	33.031	0.000	204.546	230.411
Radio	2.9735	0.235	12.644	0.000	2.512	3.435
Social Media	-0.1391	0.676	-0.206	0.837	-1.467	1.189
TV_Low	-154.5736	4.949	-31.231	0.000	-164.295	-144.852
TV_Medium	-75.5947	3.647	-20.726	0.000	-82.759	-68.431
Influencer_Mega	2.4948	3.462	0.721	0.471	-4.305	9.295
Influencer_Micro	2.9391	3.378	0.870	0.385	-3.695	9.574
Influencer_Nano	0.8015	3.346	0.240	0.811	-5.770	7.373

```

=====
Omnibus:                 58.711    Durbin-Watson:           1.874
Prob(Omnibus):            0.000    Jarque-Bera (JB):        17.802
Skew:                     0.057    Prob(JB):                 0.000136
Kurtosis:                 2.143    Cond. No.                 147.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

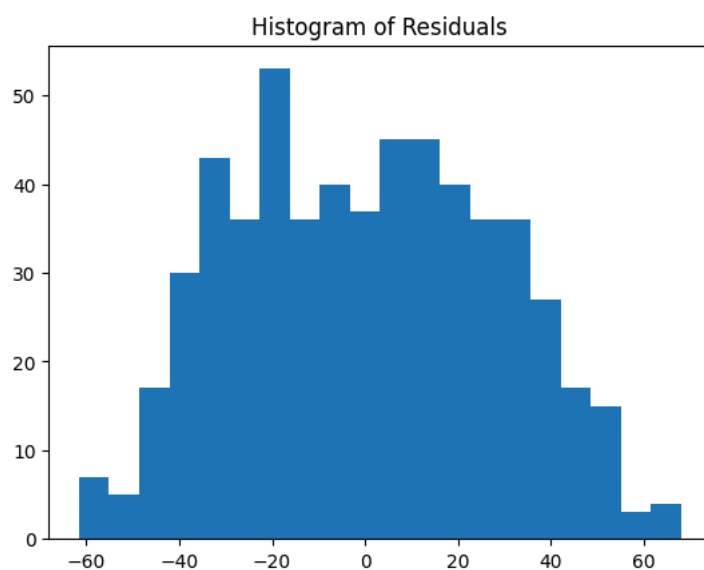
```

plt.hist(model.resid,bins=20)

plt.title("Histogram of Residuals")

plt.show()

```

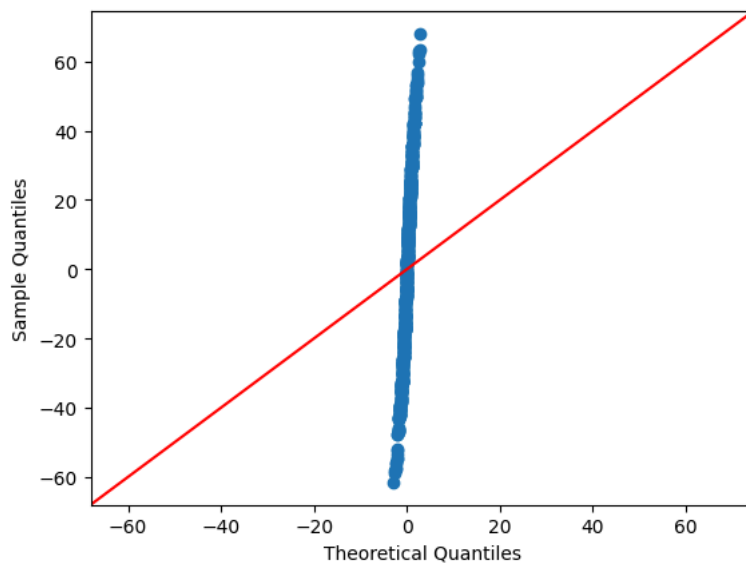


```

sm.qqplot(model.resid,line='45')

plt.show()

```



```
plt.scatter(model.fittedvalues,model.resid)

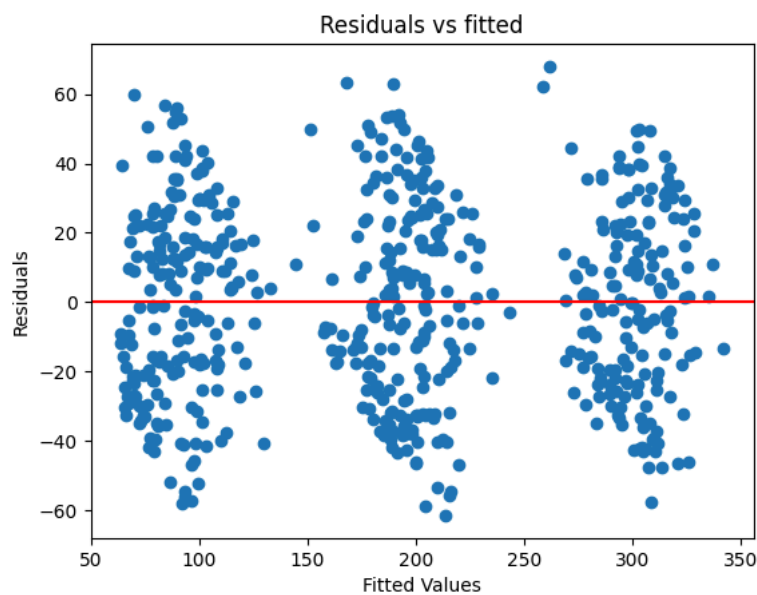
plt.axhline(0,color='red')

plt.xlabel("Fitted Values")

plt.ylabel("Residuals")

plt.title("Residuals vs fitted")

plt.show()
```



```
print(model.summary())
```

```

=====
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```

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=====
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Kurtosis:               2.143    Cond. No.                     147.
=====

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```

BUSINESS RECOMMENDATION

Based on the multiple Linear Regression analysis, the model explains approximately 90.4% of the variation in Sales ($R^2=0.904$), indicating that it provides a strong fit to the data.

The results show that Radio advertising has a statistically significant positive effect on sales ($p < 0.05$). This suggests that increasing investment in Radio advertising is likely to increase sales while holding the other marketing channels constant.

The Low TV advertising category has a significant negative effect on sales, indicating that maintaining a low level of TV advertising may reduce sales performance. Therefore, the company should avoid underinvesting in TV advertising.

The Social Media and influencer variables were not statistically significant in this model ($p > 0.05$). This suggests that their effects on sales are not clearly supported by the available data. The company should review these campaigns, improve their effectiveness, or conduct additional analysis before increasing spending in these areas.

FINAL RECOMMENDATION

1. Prioritize investment in Radio advertising, as it has the strongest positive and statistically significant relationship with Sales.
2. Avoid maintaining low levels of TV advertising, since this is associated with lower Sales.
3. Evaluate and optimize Social Media and influencer marketing strategies before allocating additional budget.
4. Continue using data-driven analysis to monitor campaign performance and improve future marketing decisions.

INTERPRETATION & EVALUATION

MODEL PERFORMANCE

The Multiple Linear Regression model achieved an R-squared of 0.904 and an Adjusted R-squared of 0.903. This indicates that approximately 90.4% of the variation in Sales is explained by the marketing variables included in the model. The high Adjusted R-squared suggests that the model provides an excellent fit to the data.

STATISTICAL SIGNIFICANCE

The overall model is statistically significant with a prob (F-statistic) of $1.82e-282$, which is far below 0.05. Therefore, the model provides strong evidence that the marketing variables collectively explain Sales.

INTERPRETATION OF COEFFICIENTS

1. Radio has a positive coefficient (2.9735) and a p-value of 0.000, indicating that Radio advertising has a statistically significant positive effect on Sales. Holding all other variables constant, increasing Radio advertising by one unit is associated with an increase of approximately 2.97 units in Sales.
2. Social Media has a coefficient of approximately -0.1391 with a p-value of 0.837, indicating that its effect is not statistically significant.
3. TV_Low has a negative coefficient (-154.57) with a P-value of 0.000, indicating that the low TV advertising category is associated with significantly lower Sales compared to the reference TV category.
4. TV_Medium has a negative coefficient (-75.59) but a p-value of 0.471, indicating that it is not statistically significant.
5. Influencer_Mega, Influencer_Micro, and influencer_Nano all have p-values greater than 0.05, suggesting that they do not have a statistically significant impact on Sales after controlling for the other variables.

CONCLUSION & BUSINESS INSIGHTS

The Multiple Linear Regression model explains approximately 90% of the variation in Sales, making it a reliable model for predicting marketing performance.

Among all predictors, Radio advertising shows the strongest statistically significant positive relationship with Sales. The analysis also indicates that Low TV advertising category is associated with substantially lower Sales compared with the reference category. The remaining TV category and influencer categories were not statistically significant in this model.

BUSINESS RECOMMENDATION

Based on this analysis, the company should prioritize investment in Radio advertising, as it has the strongest positive and statistically significant relationship with Sales. Marketing managers should review the effectiveness of social media and influencer campaigns, since they did not demonstrate statistically significant contributions in this model. Maintaining higher levels of TV advertising rather than the low category may also help improve sales outcomes.

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